

Coordination Before Collision: A Governance Framework for Multi-Agent Clinical AI

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Abstract Clinical AI systems are beginning to move beyond advice toward multi-step action in scheduling, referral management, documentation, and care coordination. When such systems operate concurrently, harms may arise not only from an individual agent's output but from interactions across shared patients, workflows, data, and finite resources. Existing healthcare standards and general-purpose agent protocols provide important foundations for data exchange, authorization, tool access, and message transport, yet they do not consistently specify how autonomous clinical actors should expose intent, honor cross-workflow dependencies, coordinate contested resources, or enter a safe mode when coordination fails. We introduce CIRC, Clinical Infrastructure for Reliable Coordination, a conceptual governance framework intended to support requirements analysis and stimulate protocol design, not a proposed standard or validated architecture. CIRC organizes a non-exhaustive taxonomy of agent-local, interaction-and-workflow, and system-and-institutional risks, and maps them to independent governance levers for identity and attribution, scope and authorization, structured intent and shared state, dependency coordination, resource awareness, operational human oversight, and monitoring and failure containment. Deployments may require different combinations of these levers; they do not form a cumulative maturity sequence. We use CliniCARE-Bench as a worked example of how trace-observable agent-local risks can be evaluated without inspecting private model reasoning, while identifying multi-agent interaction and institutional risks that current benchmarks do not test. CIRC therefore offers a starting vocabulary for connecting clinical-agent risks, controls, and evidence requirements before deployment.

1 Introduction

Clinical artificial intelligence (AI) is shifting from decision support toward systems that can select and execute multiple steps in clinical or administrative workflows.[1–3] In this Perspective, a *clinical agent* is a software system that can pursue a delegated objective by choosing and performing workflow actions with less immediate human initiation than conventional decision support. This definition includes administrative agents, supervised clinical agents, and agents permitted to initiate clinically consequential changes; these categories carry different risks and should not be governed identically.

Healthcare's dominant standards, including FHIR, SMART on FHIR, and CDS Hooks, were designed for data exchange, application authorization, and workflow-triggered decision support.[4–6] These capabilities remain essential and can support portions of agentic workflows through profiles, implementation guides, workflow engines, and local extensions. The governance question is narrower: which additional conventions may be needed when software actors operate continuously, initiate actions, and interact across organizational and vendor boundaries?

Potential failures include stale shared state, actions executed before prerequisites are satisfied, incompatible changes by different agents, retry loops, resource congestion, and escalations with no accountable recipient. We call harms arising from such dependencies *interaction risks*. The term does not imply that every multi-agent failure requires a new protocol: centralized orchestration, EHR-native workflow controls, policy engines, and human-led

processes may implement some or all of the needed safeguards.

Single-agent evaluation remains necessary but may be insufficient for systems whose actions interact. Evaluation must distinguish errors in an agent's own investigation from failures created by shared state, dependencies, hand-offs, resource competition, or institutional response. This distinction motivates a taxonomy that links each risk to observable evidence and candidate controls rather than placing deployments on a single ladder.

We introduce CIRC, Clinical Infrastructure for Reliable Coordination, as a conceptual governance framework intended to stimulate protocol design rather than a proposed standard, implemented architecture, or certification scheme. CIRC makes three contributions. First, it offers a non-exhaustive starting taxonomy spanning agent-local, interaction-and-workflow, and system-and-institutional risks. Second, it defines independent governance levers that institutions can select according to a deployment's autonomy, clinical consequences, and organizational setting. Third, it connects those risks and controls to evaluation evidence, using an existing trace-based clinical benchmark as a concrete example while making explicit which multi-agent risks remain untested. The levers are neither claimed to be minimal nor sufficient, and future extensions are expected.

2 Existing infrastructure: capabilities and remaining gaps

Healthcare has invested decades in interoperability infrastructure, including HL7v2 for clinical messaging, FHIR

for structured data and workflow resources, SMART on FHIR for application authorization, and CDS Hooks for workflow-triggered decision support.[4–8] These standards already provide more of the substrate for agentic workflows than a simple “data exchange” characterization suggests. FHIR Provenance and AuditEvent can record actors, activities, policies, and outcomes. FHIR Task represents work state, owners, inputs, outputs, composite tasks, and links to formal workflow definitions; the specification explicitly accommodates both distributed task management and central workflow engines. Subscription supports event notification. SMART Backend Services authorizes headless applications, and CDS Hooks invokes services at defined clinical workflow events.

The broader AI ecosystem also provides substantial coordination machinery. A2A defines agent cards, signed capability descriptions, stateful tasks, task cancellation, streaming and push updates, authorization states, extensions, and security guidance.[9, 10] MCP defines a host-controlled architecture for connecting models to isolated tool and resource servers, with capability negotiation and host-enforced security boundaries.[11, 12] ACP-style protocols similarly support task and message exchange.[13] The open question is therefore not whether these technologies can participate in clinical coordination. It is which healthcare-specific semantics and conformance expectations should be shared across implementations.

Table 1 distinguishes capabilities already present from capabilities that are representable through profiles, extensions, or local infrastructure, and from governance contracts that are not yet consistently standardized. The boundary is not absolute. A health system can implement many CIRC levers today without waiting for a new standard; CIRC’s proposed contribution is to connect those implementation choices to explicit risk and evidence requirements.

Existing AI risk frameworks characterize what can go wrong. The AI Risk Repository synthesizes more than 700 risks across 43 taxonomies, and the NIST AI Risk Management Framework organizes governance functions for trustworthy systems.[14, 15] Clinical agent benchmarks evaluate whether individual agents perform correctly in realistic environments.[16, 17] Multi-agent clinical deployment additionally raises questions about interacting agents acting concurrently on shared patients, finite resources, and institutional workflows. Four gaps illustrate this distinction.

Inconsistent agent observability and attribution

As agents move from recommending actions to initiating orders, updating documentation, routing referrals, or modifying schedules, audit infrastructure should distinguish human-authored actions, human-approved agent actions, and autonomous agent-initiated actions. FHIR Provenance and AuditEvent can represent software actors, activities, policies, outcomes, and signatures. The remaining issue is implementation consistency: deployments would need profiles and policies requiring stable agent and model versions,

delegation context, and links between the clinical action and its audit record. Without that consistency, adverse-event review may still struggle to reconstruct which agent acted, under whose authority, and within which scope.

Unstandardized clinical intent and dependency semantics

Clinical coordination requires expressing dependencies across disease-focused, patient-focused, capability-focused, and operational agents. FHIR Task, RequestGroup, CarePlan, and workflow definitions can represent much of this structure, while A2A tasks and extensions can carry it across agents. The gap is a shared clinical profile specifying which prerequisites, timing windows, state versions, resource needs, escalation rules, and reversal obligations must be declared for a given class of autonomous action. Such a profile could extend existing standards rather than replace them.

Authorization without a shared model of clinical autonomy

Healthcare defines role-based authority around expertise, licensure, institutional policy, and accountability. SMART scopes and A2A authorization mechanisms can authenticate applications, limit data access, and require additional authorization. Their policies are intentionally implementation specific. A clinical deployment still must define whether an agent may read, recommend, draft, place, or execute an action; for which patients and settings; and with which approval and escalation path. CIRC calls for these distinctions to be machine-readable and auditable, not for a replacement of OAuth, SMART, or institutional access-control systems.

Locally implemented system-level resource coordination

When many agents independently conclude that the same time window is optimal for imaging, infusion, discharge planning, or specialist review, they could collectively overwhelm shared resources even if each decision is locally reasonable. Prior work calls this type of convergence a “digital stampede.”[18] Historical utilization statistics and single-agent benchmarks may not expose risks arising dynamically from the interaction of agents, patients, resources, and workflows. Existing standards can represent many relevant facts, and workflow engines can coordinate specified processes, but aggregate demand signals and population-level stress tests are not routinely portable across clinical agent ecosystems. This is chiefly an implementation and governance gap: institutions may reasonably keep resource arbitration centralized rather than expose it through a peer-to-peer protocol.

The Ripple Effect Protocol (REP) motivates this population-level failure pattern but is not evidence that CIRC is effective in healthcare. REP exchanges sensitivity signals among agents and reports results from nonclinical simulated domains; CIRC neither requires that mechanism nor assumes those results transfer to clinical operations. Aggregate demand or sensitivity signals are one candidate implementation of resource awareness, and would require

Table 1: Existing infrastructure and the narrower coordination questions raised by CIRC.

Standard or approach	Capabilities already provided	Extension or governance work for clinical-agent use
FHIR resources	Structured clinical data; Task lifecycle, ownership, inputs, outputs, and composite work; Provenance and AuditEvent; event subscriptions.	Semantics and conformance: profiles can encode agent versions, dependencies, and state, but common clinical-agent profiles, required fields, and cross-vendor behavior would need agreement.
SMART on FHIR	OAuth-based authorization for user-facing and headless clients; patient and encounter context; resource-level scopes and token introspection.	Authorization policy: local policy can restrict actions, but patient-specific authority, degree of autonomy, approval conditions, and revocation across agents require explicit policy and enforcement.
CDS Hooks	Synchronous invocation at named clinical workflow events; context and prefetch; cards, suggestions, and SMART app launch.	Workflow: useful for human-mediated checkpoints, but not a durable multi-agent dependency graph, transaction protocol, or population resource controller.
A2A and ACP-style protocols	Agent discovery, capability advertisement, messages, stateful tasks, status updates, cancellation, authentication patterns, and extensions.	Clinical governance: healthcare intent, patient and encounter binding, action authority, clinical prerequisites, accountable escalation, and safe-mode semantics require domain profiles or local rules.
MCP	Host-mediated access to tools, resources, and prompts; capability negotiation; isolation of server connections; host control of permissions and consent.	Scope: MCP can expose governed clinical tools, but it is not a peer-agent workflow or shared clinical dependency standard.
Workflow and policy engines	Stateful orchestration, business rules, human tasks, access decisions, retries, timeouts, and compensation within a governed deployment.	Implementation and portability: these may implement CIRC levers locally; cross-vendor semantics, evidence exchange, and institutional trust remain deployment-specific unless profiled.

separate evaluation of privacy, manipulation, allocation fairness, workflow effects, and clinical safety.

3 A non-exhaustive taxonomy of clinical-agent risk

CIRC begins with risks rather than levels. The taxonomy is intentionally non-exhaustive: it is a starting point for requirements analysis, not a claim to enumerate every hazard or rank deployments on a maturity scale. We organize risks by the locus at which evidence and control are needed (Table 2). A single event can occupy more than one domain, and risk domains do not imply causal independence.

The first domain is already visible in emerging clinical-agent evaluations. CliniciCARE-Bench, for example, evaluates retrospective clinical audits over 25 scenarios and 750 patient-specific cases in a governed, replayable EHR environment.[19] It separately scores final verdicts, patient-evidence grounding, policy use, process adherence, and calibrated abstention. Across 16 evaluated systems, reported four-way accuracy ranged from 65.3% to 76.1%, while accuracy that also excluded prohibited process shortcuts was 4.8 to 14.8 percentage points lower. Every evaluated system over-committed more often than it over-abstained. These findings do not validate CIRC and do not test multi-agent coordination. They demonstrate something narrower and useful: governance-relevant properties can be specified against observable traces, and a correct output can conceal a defective investigation.

Interaction and system risks require different evidence. A benchmark in which one agent reviews one record cannot measure cross-agent deadlock, conflicting modifications, population-level demand, or whether a clinician responds to an escalation. These risks require multi-agent replay, workflow stress tests, sociotechnical simulation, prospective shadow deployment, or post-deployment monitoring. CIRC

uses the taxonomy to make those evidence boundaries explicit rather than treating single-agent accuracy as a proxy for system safety.

4 CIRC governance levers

CIRC maps risks to independent governance levers (Table 3). A lever is a capability or institutional control that can be implemented through a protocol, a centralized service, an EHR-native workflow, or another architecture. Deployments may implement the levers in different orders and at different strengths. None is asserted to be individually necessary, collectively sufficient, or unique to CIRC.

The mapping is many-to-many. Dependency coordination can mitigate both agent-local omissions and cross-agent prerequisite violations. Identity supports auditability but does not, by itself, prevent an authorized agent from acting on stale state. Human oversight can contain uncertainty or conflict only when escalation has an accountable recipient, response window, and fallback behavior. Resource signals may reduce congestion while introducing privacy, manipulation, and allocation risks of their own.

This framing is compatible with multiple architectures. A central workflow engine may maintain the dependency graph; an institutional policy engine may enforce scope; an EHR-native service may mediate orders; a supervisory agent may detect conflict; and direct agent-to-agent messages may carry intent across vendors. Protocol-level fields are most useful where information must cross those boundaries, but CIRC does not assume that every control belongs in a new protocol or that decentralized agent communication is preferable to orchestration.

Table 2: A non-exhaustive taxonomy of clinical-agent risks.

Risk domain	Illustrative risks	Primary unit of evaluation
Agent-local evidence and reasoning	Wrong-patient or wrong-episode attribution; temporal misalignment; substitution of codes or orders for clinical evidence; incomplete evidence retrieval; rule misapplication; unsupported inference; false closure under missing or conflicting evidence.	One agent’s observable investigation, actions, report, and abstention behavior.
Interaction and workflow	Stale or inconsistent shared state; unmet prerequisites; incompatible or duplicated actions; failed handoffs; race conditions; incorrect dependency graphs; recursive retry or negotiation loops.	Concurrent trajectories around a shared patient, task, or care pathway.
System and institutional	Resource congestion; correlated failure across patients; unavailable or misleading aggregate signals; escalation without an accountable responder; coordination-layer outage; cross-organizational trust failure; alert burden, inequity, and drift.	Agent populations, shared resources, institutions, and human response processes over time.

Table 3: Independent CIRC governance levers and the questions they make answerable.

Governance lever	Core question	Illustrative evidence
Identity and attribution	Which agent and version acted, under whose authority, and in which execution context?	Signed identity, delegation chain, provenance record, immutable audit event.
Scope and authorization	Was this action permitted for this patient, task, setting, and degree of autonomy?	Machine-readable capability and scope declaration; policy decision and override record.
Structured intent and shared state	What action is proposed, against which state version, with what time limit and reversal plan?	Typed intent message, state version, expected effect, expiry, and idempotency key.
Dependency coordination	Which prerequisites and downstream obligations must hold before and after the action?	Dependency graph, prerequisite checks, acknowledgement, cancellation propagation.
Resource and population awareness	Could locally reasonable actions create congestion or correlated failure?	Privacy-preserving capacity, queue, demand, priority, and concentration signals.
Operational human oversight	Who must respond, by when, with what authority, and what happens after non-response?	Named role, response deadline, fallback recipient, override, suspension, and closure.
Monitoring and failure containment	How are faults detected, bounded, revoked, and handled during degraded operation?	Health state, replay protection, rate limits, circuit breakers, safe mode, and recovery log.

Risk locus	Governance task	Candidate levers
Agent-local	Establish whether evidence, timing, policy, scope, and uncertainty support an action.	Identity; scope; structured state; oversight
Interaction and workflow	Preserve shared state and dependencies as actions cross agents and systems.	Structured intent; dependency coordination; containment
System and institutional	Detect correlated demand and maintain accountability during escalation or outage.	Resource awareness; oversight; monitoring and containment

Figure 1: CIRC connects risk loci to governance tasks and candidate controls. Arrows are intentionally not one-to-one: controls overlap, and the appropriate implementation depends on the deployment.

4.1 Operational definitions and a coordination message

The framework becomes testable only when its terms can be recognized in system behavior. We use the following operational definitions. An *interaction risk* is a potential or realized failure attributable to interdependence among multiple actors, actions, shared state, or resources rather than to one output considered in isolation. A *collision* occurs when two or more actions target the same state or resource during overlapping validity windows and violate a declared compatibility rule. A *scope violation* is an attempted or executed action outside the authorization valid for the actor, patient, task, setting, time, and delegated purpose at the decision point. A *dependency violation* occurs when an action passes its release gate while a declared prerequisite is absent, failed, expired, or based on an incompatible state version. *Failure containment* means that a detected fault is held, quarantined, compensated, or reversed before it crosses a predefined patient, workflow, resource, or institutional boundary.

Listing 1 gives one concrete coordination message for an imaging-scheduling agent proposing to move an appointment. The example is illustrative rather than a normative CIRC wire format. The same fields could be represented through a FHIR profile, an A2A extension, or an institutional workflow schema. Patient and encounter references are placeholders, not clinical data.

Listing 1: Illustrative structured coordination message.

```
{
  "schema": "circ.intent.v0",
  "message_id": "urn:uuid:example-message",
  "idempotency_key": "schedule-change-example-1",
  "actor": {
    "agent_id": "agent:scheduling:example",
    "version": "3.2.1",
    "issuer": "Organization/institution-a"
  },
  "authority": {
    "action": "schedule.modify",
    "delegated_by": "PractitionerRole/care-team",
    "policy": "Policy/outpatient-scheduling-v4"
  },
  "subject": {
    "patient": "Patient/example",
    "encounter": "Encounter/example"
  },
  "state": {
    "version": "W/example-etag",
    "observed_at": "2030-01-15T14:30:00Z"
  },
  "intent": {
    "proposed_action": "reschedule imaging",
    "target": "Task/imaging-example",
    "reason": "capacity adjustment"
  },
  "dependencies": [
    { "task": "Task/oncology-clearance", "required_state": "completed" },
    { "task": "Task/observation-visit", "constraint": "must_remain_scheduled" }
  ],
  "valid_until": "2030-01-15T15:00:00Z",
  "reversal": { "action": "restore_prior_slot", "deadline_minutes": 30 },
  "escalation": {
    "owner_role": "oncology attending",
    "respond_by": "2030-01-15T14:50:00Z",
    "on_timeout": "hold"
  },
  "signature": "detached-jws:example"
}
```

The receiver can evaluate this proposal without inferring critical context from free text: it can authenticate the actor, evaluate action-specific authority, compare state versions, inspect prerequisites, deduplicate retries, enforce expiry, and route unresolved conflict to an accountable role. A valid message does not imply a valid clinical decision; it makes the decision context inspectable and enforceable.

4.2 One possible deployment architecture

Figure 2 shows one centralized realization of the levers. Agents communicate through a coordination service or workflow engine, which checks institutional identity and policy, binds actions to EHR state, appends audit evidence, and routes escalations to a named human queue. An external agent crosses an institutional boundary with a signed, scoped intent message and remains subject to local policy. The diagram is not a required topology: the same interfaces may be distributed, embedded in the EHR, or divided among existing services.

4.3 Implementation patterns and trade-offs

The same governance lever can be realized through different architectures (Table 4). The appropriate choice depends on institutional boundaries, workflow stability, latency, reversibility, and the consequences of coordinator failure.

CIRC is therefore best understood as an architecture-neutral requirements framework. It asks whether the chosen design can provide each needed control and its evidence trail. A FHIR implementation guide, an A2A clinical extension, or a local orchestration policy could each satisfy portions of that contract. A new protocol primitive is warranted only where existing representations cannot express a required concept or where cross-system conformance cannot otherwise be achieved.

5 Failure model and degraded operation

A coordination layer is itself safety-critical infrastructure and can create common-mode failure. CIRC therefore assumes that agents, credentials, messages, state stores, dependency models, coordinators, and human response processes can each fail or be compromised. It does not assume that failures are independent or benign. The framework does not prescribe a universal identity issuer or consistency algorithm; it requires each deployment to declare its trust anchors, revocation mechanism, state guarantees, and degraded behavior. This follows a zero-trust posture in which access is evaluated for the requesting subject, resource, and context rather than inferred from network location.[20]

Table 5 states the principal failure surfaces. The controls are illustrative: their adequacy depends on clinical consequences, timing, reversibility, and organizational boundaries. Classical state-machine replication can improve availability and consistency, for example, but does not make an incorrect dependency model clinically correct.[21]

Safe mode should be defined by action class rather than by one universal “fail closed” rule. Read-only retrieval or analysis may continue during partial outage only when

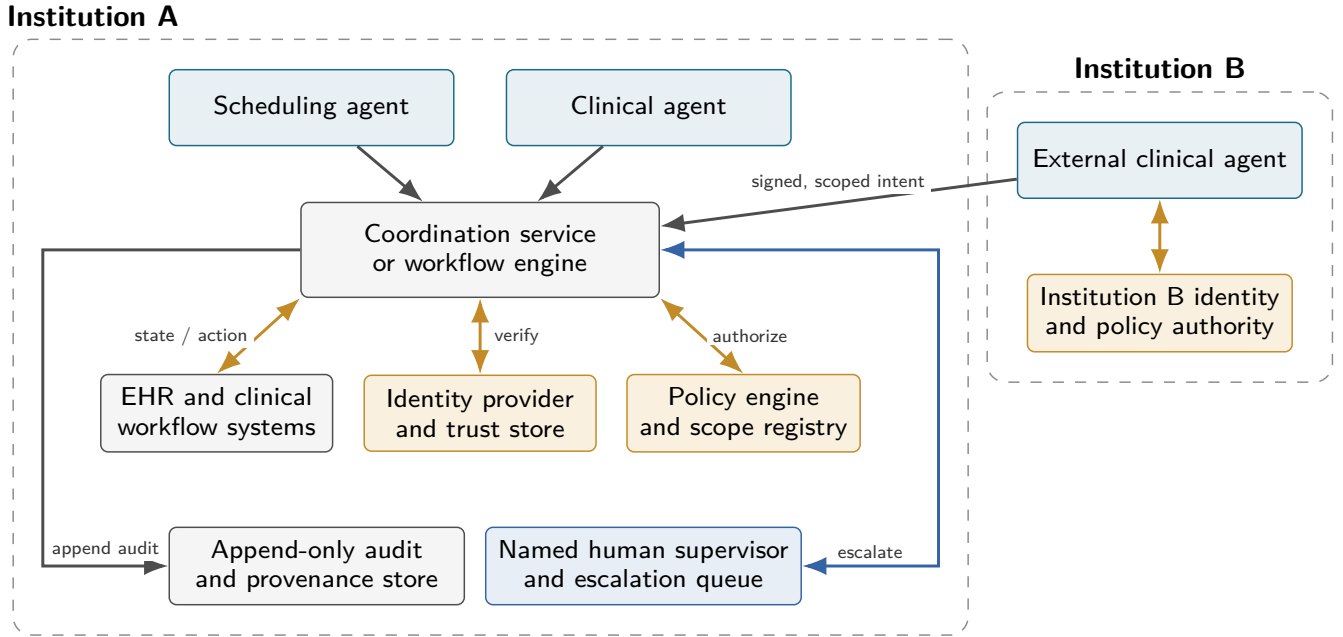


Figure 2: One possible implementation of CIRC governance levels. Identity, authorization, clinical state, audit, and human escalation remain institutionally governed even when an external agent uses a shared coordination message. The coordination service may be replaced by an EHR-native workflow or another architecture that provides equivalent controls and evidence.

Table 4: Alternative architectures for implementing CIRC governance levels.

Pattern	Principal strengths	Limitations and role of shared conventions
Central workflow orchestrator	One authority for sequencing, retries, compensation, and human tasks; clear operational ownership.	Can become a bottleneck or single failure domain; cross-vendor inputs still need stable identity, intent, state, and outcome semantics.
EHR-native workflow control	Close to orders, patient context, existing permissions, and clinician work queues.	May be vendor- or institution-specific and difficult to extend across external agents or non-EHR systems.
Policy engine and identity layer	Centralized authorization, revocation, separation of duties, and consistent policy decisions.	Does not by itself coordinate task state, clinical prerequisites, resource contention, or human response.
Supervisory agent	Can synthesize context and detect conflicts without requiring every peer to communicate directly.	Introduces another model failure surface; requires bounded authority, independent monitoring, and deterministic enforcement for consequential actions.
Distributed agent protocol	Supports cross-vendor discovery, delegation, and task exchange without one workflow owner.	Increases consistency, trust, conflict-resolution, and outage-recovery demands; clinical profiles are needed above the transport.
Human-led coordination	Uses existing professional accountability and adapts to unusual cases.	Limited by latency, workload, alert burden, and diffusion of responsibility; agent escalations still need named recipients and closure rules.

Table 5: Failure model for the clinical-agent coordination layer.

Failure surface	Illustrative failures	Required containment behavior
Identity and trust	Compromised agent; false scope declaration; credential theft; stale or unrevoked credential; untrusted external issuer.	Verify issuer, audience, version, delegation, and revocation at decision time; reject or quarantine the actor; preserve evidence for incident review.
State and messaging	Stale patient state; loss, duplication, reordering, or replay; conflicting versions; delayed acknowledgement; network partition.	Bind actions to state versions; use idempotency and replay protection; expose uncertainty; prevent irreversible execution on unresolved state.
Dependency and workflow	Missing or incorrect prerequisite; cyclic dependency; deadlock or livelock; unbounded retry; partial cancellation or compensation.	Validate graphs, bound retries, detect cycles, propagate cancellation, and freeze the affected pathway when consistency cannot be restored.
Arbitration and resource signals	Incorrect priority; manipulated or stale capacity; starvation; correlated demand; denial of service.	Authenticate and timestamp aggregate signals, enforce rate and fairness policies, detect concentration, and transfer allocation to a designated fallback.
Coordination service	Outage, corrupted state, faulty arbitration, software update failure, or loss of quorum.	Use health checks and redundancy appropriate to risk; expose degraded status; restore from auditable checkpoints; avoid silent continuation under unknown control state.
Human control path	Alert not delivered or acknowledged; wrong recipient; alert overload; automation bias; conflicting overrides; diffusion of responsibility.	Require named ownership, deadline and acknowledgement, fallback escalation, override authority, workload monitoring, and documented closure.

data freshness and missing coordination context are visible to the user. Reversible administrative actions may be queued, deduplicated, or released after explicit confirmation. Actions that change treatment, orders, medication, disposition, or other consequential clinical state should generally be held when identity, scope, prerequisite, or shared-state checks are unavailable. Emergency exceptions require a predefined institutional pathway; the coordination layer should not invent one at runtime. In every case, the system should identify which guarantees remain active, return control to a named role, and record how normal operation was restored.

Cross-organizational identity requires federation rather than one global CIRC authority. Each institution can issue or recognize credentials under its own policy, while bilateral or network trust agreements define accepted issuers, assurance levels, claims, and revocation expectations. A valid identity establishes attribution, not clinical authority: authorization remains specific to the patient, action, setting, time, and delegated purpose.

6 Operational human oversight

“Human review” is not a control unless it changes who is responsible for what happens next. Health IT safety is sociotechnical: performance depends on people, tasks, technology, organizations, and the environment in which they interact.[22, 23] Automation can also redistribute attention and authority rather than simply reduce workload.[24] CIRC therefore treats oversight as an operational contract, not an endpoint label.

Every escalation should specify: (1) the accountable role and current assignee; (2) the triggering condition and supporting evidence; (3) the decision or action requested; (4) urgency and a response deadline; (5) whether the proposed action is blocked, queued, or continuing; (6) the fallback recipient and behavior after non-response; and (7)

the event that closes the escalation. Assignment, delivery, acknowledgement, decision, override, and closure should be distinct auditable states.

Institutions must also define governance outside the immediate alert. A designated body should approve agent scopes and material changes; operational owners should be able to pause an agent; credential and policy administrators should revoke access; and a clinical authority should resolve conflicting recommendations. Separation of duties is important when one vendor supplies both the acting agent and the supervisory component. Escalation policies should suppress duplicates, group related concerns, set alert budgets, and monitor acknowledgement and override patterns so that adding a coordination layer does not simply add another source of alert fatigue.

Useful oversight measures include time to assignment, acknowledgement, and decision; fraction of escalations with no response by deadline; fallback activation; duplicate-alert rate; override and reversal rate; unresolved workload per accountable role; and the fraction of consequential actions executed without documented closure. These measures evaluate the response process rather than merely counting alerts.

7 Regulatory and institutional positioning

Regulatory status attaches to a software function, its intended use, users, claims, and deployment context—not to the labels “agent” or “CIRC” and not to a centralized or distributed topology. In the United States, FDA guidance distinguishes clinical decision support functions that satisfy the statutory non-device criteria from software functions that remain devices.[25] A multi-agent deployment may contain administrative functions, non-device decision support, and device software functions at the same time. Each function therefore requires its own assessment, including whether combining it with autonomous execution or other

Table 6: Boundary between CIRC evidence and regulatory or institutional decisions.

Decision layer	Question that must be answered	Role of CIRC
Classification	What is each function’s intended use, and which legal category and actor roles apply?	Can inventory functions, autonomy, actions, and responsible organizations; does not assign legal status.
Premarket or conformity assessment	Is there adequate evidence of safety, performance, and clinical benefit for the intended use?	Can expose control operation and trace provenance; does not replace analytical, clinical, usability, or system validation.
Institutional deployment	Who may activate, authorize, monitor, pause, and respond to the system in this workflow?	Can encode local scope, accountability, escalation, and safe mode; does not transfer professional or organizational responsibility.
Change and post-market control	Which modifications are allowed, what requires reassessment, and how are emerging risks detected?	Can bind versions to impact assessments and monitoring evidence; does not authorize a modification or define the reporting threshold.

agents changes its intended use or risk.

For an AI-enabled device software function, an authorized predetermined change control plan can describe specified future modifications, the method for developing, validating, and implementing them, and the associated impact assessment.[26] CIRC records such as stable versions, delegation, policy decisions, dependency checks, reversals, and post-deployment events could support that lifecycle evidence. They do not establish safety or effectiveness, expand the authorized intended use, or make an out-of-plan modification permissible.

The European Union AI Act likewise bases obligations on intended purpose, statutory role, and classification. For systems classified as high-risk, its requirements include risk management, record keeping, human oversight, accuracy, robustness, cybersecurity, quality management, and post-market monitoring, with different duties for providers and deployers.[27] CIRC levers overlap with several of these operational concerns, but implementing a lever is not itself a demonstration of conformity. Privacy, cybersecurity, medical-device, professional-practice, and health-system obligations also remain independently applicable.

Before release, a deployment dossier should therefore map every consequential action to the function that proposes and executes it, its intended use and autonomy, the responsible legal and operational actors, applicable authorization, validation evidence, dependencies, affected populations, and fallback behavior. Changes to a model or agent version, permitted action, dependency, trust boundary, patient population, resource policy, or human fallback should trigger a recorded impact assessment proportional to the risk. This is an institutional change-control rule, not a claim that every change requires a new regulatory submission.

8 From risk taxonomy to evaluation

The taxonomy separates three questions that were conflated in the submitted simulations. First, can a risk be observed? Candidate measures include attribution completeness, stale-state reads, unresolved dependencies, duplicated or conflicting actions, retry amplification, queue concentration, safe-mode activation, and unowned or overdue escalations. Second, can a control detect or contain

the risk without blocking valid work? Matched stress tests should report both fault containment and preservation of unaffected workflows. Third, does the control improve clinical or operational outcomes? That question requires prospective evaluation and cannot be inferred from a deterministic demonstration.

Trace-based benchmarks such as CliniCARE provide a concrete template for the first question. They log retrieval, computation, policy access, citations, and reports, allowing process defects to be separated from final-answer errors without inspecting private reasoning. Extending this approach to interaction risks would require linked traces from multiple agents acting on shared state, together with pre-specified fault models, applicable denominators, baselines, repeated runs, uncertainty estimates, and failure cases. Population-level resource tests would additionally need realistic arrival processes, capacity constraints, and human response models.

Accordingly, this Perspective does not present the removed toy simulations as evidence of control effectiveness. Instead, it proposes a staged evaluation agenda: trace-based assessment of agent-local risks; matched multi-agent replay for interaction and workflow risks; system stress testing for shared resources and common-mode failure; prospective shadow deployment; and monitored clinical use where regulatory and institutional requirements permit.

9 Limitations and future directions

CIRC is a conceptual and non-exhaustive framework. It does not specify a complete technical implementation, establish the necessity or sufficiency of any lever, certify deployment readiness, or demonstrate clinical benefit. The taxonomy will require extension as clinical agents acquire new capabilities and as evidence emerges from real workflows. Its categories also overlap: a stale-state error, for example, can reflect an agent-local retrieval failure, an interaction failure, and an institutional monitoring failure.

CliniCARE illustrates trace-observable agent-local risks, but it is a retrospective single-agent audit benchmark rather than a test of multi-agent coordination. Its findings cannot support claims about cross-vendor messaging, shared-resource control, human response, or prevented pa-

tient harm. Likewise, population-level clinical prediction is outside the scope of a coordination framework and must be developed and validated as a separate clinical model.

The threat model and oversight contract identify requirements but do not establish that any proposed implementation will satisfy them. Choices about trusted issuers, consistency, emergency exceptions, response deadlines, and safe-mode behavior remain institution- and action-specific. Governance questions also include liability, local configuration, alert burden, and equitable access. The regulatory discussion above is descriptive rather than legal guidance; applicability depends on the specific function, intended use, jurisdiction, and organizational role. Future work should compare protocol, orchestration, policy-engine, EHR-native, and human-led implementations against common risk-specific endpoints.

10 Conclusion

Multi-agent clinical safety cannot be inferred from individual-agent performance alone. The relevant risks span local investigations, interactions across shared workflows, and the institutions that authorize, monitor, and respond to autonomous action. CIRC offers a starting taxonomy for those risks and a set of independent governance levers that can be implemented through different technical architectures.

The framework’s value should be judged by whether it makes requirements and evidence more precise, not by treating its categories as a maturity ladder or its controls as established safety invariants. Existing trace-based benchmarks show how some agent-local risks can be measured. The next challenge is to build equally inspectable evaluations for shared state, dependencies, resource competition, degraded operation, and human response before autonomous clinical agents operate at scale.

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References

- [1] Michael Moritz, Eric Topol, and Pranav Rajpurkar. Coordinated ai agents for advancing healthcare. *Nature Biomedical Engineering*, 9:432–438, 2025. doi: 10.1038/s41551-025-01363-2.
- [2] Andrew A. Borkowski and Alon Ben-Ari. Multiagent ai systems in health care: Envisioning next-generation intelligence. *Federal Practitioner*, 42:188–194, 2025. doi: 10.12788/fp.0589.
- [3] R. Andrew Taylor. Ai agents, automaticity, and value alignment in health care. *NEJM AI*, 2(8), 2025. doi: 10.1056/AIip2401165.
- [4] HL7 International. Fast healthcare interoperability resources (fhir), release r4, 2019. URL <https://www.hl7.org/fhir/>.
- [5] HL7 International. Cds hooks specification, release 1.0, 2018. URL <https://cds-hooks.hl7.org>.
- [6] Joshua C. Mandel, David A. Kreda, Kenneth D. Mandl, Isaac S. Kohane, and Rachel B. Ramoni. Smart on fhir: a standards-based, interoperable apps platform for electronic health records. *Journal of the American Medical Informatics Association*, 23:899–908, 2016. doi: 10.1093/jamia/ocv189.
- [7] Health Level Seven International. HL7 version 2.5.1: Application protocol for electronic data exchange in healthcare environments, 2007. URL https://www.hl7.org/implement/standards/product_brief.cfm?product_id=144.
- [8] HL7 International. Smart app launch implementation guide, version 2.2.0, 2024. URL <https://hl7.org/fhir/smart-app-launch/>.
- [9] Rao Surapaneni, Mukul Jha, Matt Vakoc, and Todd Segal. Announcing the agent2agent protocol, 2025. URL <https://developers.googleblog.com/en/a2a-a-new-era-of-agent-interoperability/>.
- [10] A2A Protocol Working Group. Agent2agent protocol specification, version 1.0.0, 2026. URL <https://a2a-protocol.org/latest/specification/>.
- [11] Anthropic. Introducing the model context protocol, 2024. URL <https://www.anthropic.com/news/model-context-protocol>.
- [12] Model Context Protocol Contributors. Model context protocol architecture, 2025. URL <https://modelcontextprotocol.io/specification/2025-06-18/architecture>.
- [13] Samuel Besen and Anna Gutowska. What is agent communication protocol?, 2025. URL <https://www.ibm.com/think/topics/agent-communication-protocol>.
- [14] Peter Slattery et al. The ai risk repository: A meta-review, database, and taxonomy of risks from artificial intelligence. *arXiv preprint arXiv:2408.12622*, 2024. doi: 10.48550/arXiv.2408.12622.
- [15] National Institute of Standards and Technology. Artificial intelligence risk management framework (ai rmf 1.0). Technical Report NIST AI 100-1, NIST, 2023.
- [16] Yixing Jiang et al. Medagentbench: A virtual ehr environment to benchmark medical llm agents. *NEJM AI*, 2(9), 2025. doi: 10.1056/AIdbp2500144.
- [17] Dyke Ferber et al. Development and validation of an autonomous artificial intelligence agent for clinical decision-making in oncology. *Nature Cancer*, 6:1337–1349, 2025. doi: 10.1038/s43018-025-00991-6.

- [18] Ayush Chopra et al. Ripple effect protocol: Coordinating agent populations. *arXiv preprint arXiv:2510.16572*, 2025. doi: 10.48550/arXiv.2510.16572.
- [19] Veronica Chatrath, Bryan Zhu, George Pu, et al. Clinicare-bench: Clinical calibrated audit of medical reasoning in ehr. Scale AI Research, 2026. URL <https://labs.scale.com/papers/clinicare-bench>.
- [20] Scott Rose, Oliver Borchert, Stu Mitchell, and Sean Connelly. Zero trust architecture. Technical Report NIST SP 800-207, National Institute of Standards and Technology, 2020.
- [21] Fred B. Schneider. Implementing fault-tolerant services using the state machine approach: A tutorial. *ACM Computing Surveys*, 22(4):299–319, 1990. doi: 10.1145/98163.98167.
- [22] Dean F. Sittig and Hardeep Singh. A new sociotechnical model for studying health information technology in complex adaptive healthcare systems. *Quality and Safety in Health Care*, 19(Supplement 3):i68–i74, 2010. doi: 10.1136/qshc.2010.042085.
- [23] Pascale Carayon, Ayse Schoofs Hundt, Ben-Tzion Karsh, Ayse P. Gurses, Christian J. Alvarado, Michelle Smith, and Patricia Flatley Brennan. Work system design for patient safety: the seips model. *Quality and Safety in Health Care*, 15(Supplement 1):i50–i58, 2006. doi: 10.1136/qshc.2005.015842.
- [24] Raja Parasuraman, Thomas B. Sheridan, and Christopher D. Wickens. A model for types and levels of human interaction with automation. *IEEE Transactions on Systems, Man, and Cybernetics Part A*, 30(3):286–297, 2000. doi: 10.1109/3468.844354.
- [25] U.S. Food and Drug Administration. Clinical decision support software: Guidance for industry and food and drug administration staff. Technical report, FDA, January 2026. URL <https://www.fda.gov/media/109618/download>. Final guidance.
- [26] U.S. Food and Drug Administration. Marketing submission recommendations for a predetermined change control plan for artificial intelligence-enabled device software functions: Guidance for industry and food and drug administration staff. Technical report, FDA, August 2025. URL <https://www.fda.gov/media/166704/download>. Final guidance.
- [27] European Parliament and Council of the European Union. Regulation (eu) 2024/1689 laying down harmonised rules on artificial intelligence, 2024. URL <https://eur-lex.europa.eu/eli/reg/2024/1689/oj/eng>.
- [28] J. Y. Kim, S. J. Nam, J. E. Lee, et al. Real-world evidence of neoadjuvant docetaxel/carboplatin/trastuzumab/pertuzumab in patients with her2-positive early or locally advanced breast cancer. *Cancer Research and Treatment*, 54:1091–1098, 2022. doi: 10.4143/crt.2021.901.
- [29] C. Hockaday, J. Chiramel, et al. Efficacy and safety of neoadjuvant docetaxel, carboplatin, trastuzumab and pertuzumab in patients with her2-positive early breast cancer. *Clinical Oncology*, 2024. doi: 10.1016/j.clon.2024.103717.
- [30] J. A. W. Polderman, J. Hermanides, and A. H. Hulst. Update on the perioperative management of diabetes mellitus. *BJA Education*, 24:261–269, 2024. doi: 10.1016/j.bjae.2024.04.007.
- [31] Centre for Perioperative Care. Guideline for the perioperative care of people with diabetes mellitus undergoing elective and emergency surgery, 2022. URL <https://cpoc.org.uk/guidelines-resources/guidelines>.
- [32] Alexander R. Lyon et al. 2022 esc guidelines on cardio-oncology. *European Heart Journal – Cardiovascular Imaging*, 23:e333–e465, 2022. doi: 10.1093/ehjci/jeac106.
- [33] Alana F. Yu et al. Cardiac safety of reduced cardiotoxicity surveillance during her2-targeted therapy. *JACC: CardioOncology*, 7:430–441, 2025. doi: 10.1016/j.jacc.2025.05.006.
- [34] David J. Kuter. Treatment of chemotherapy-induced thrombocytopenia in patients with non-hematologic malignancies. *Haematologica*, 107:1243–1263, 2022. doi: 10.3324/haematol.2021.279512.
- [35] U.S. Food and Drug Administration. Artificial intelligence/machine learning-based software as a medical device action plan. Technical report, FDA, 2021. URL <https://www.fda.gov/media/145022/download>.
- [36] Object Management Group. Business process model and notation, version 2.0.2. Technical Report formal/2013-12-09, Object Management Group, 2014. URL <https://www.omg.org/spec/BPMN/2.0.2/>.